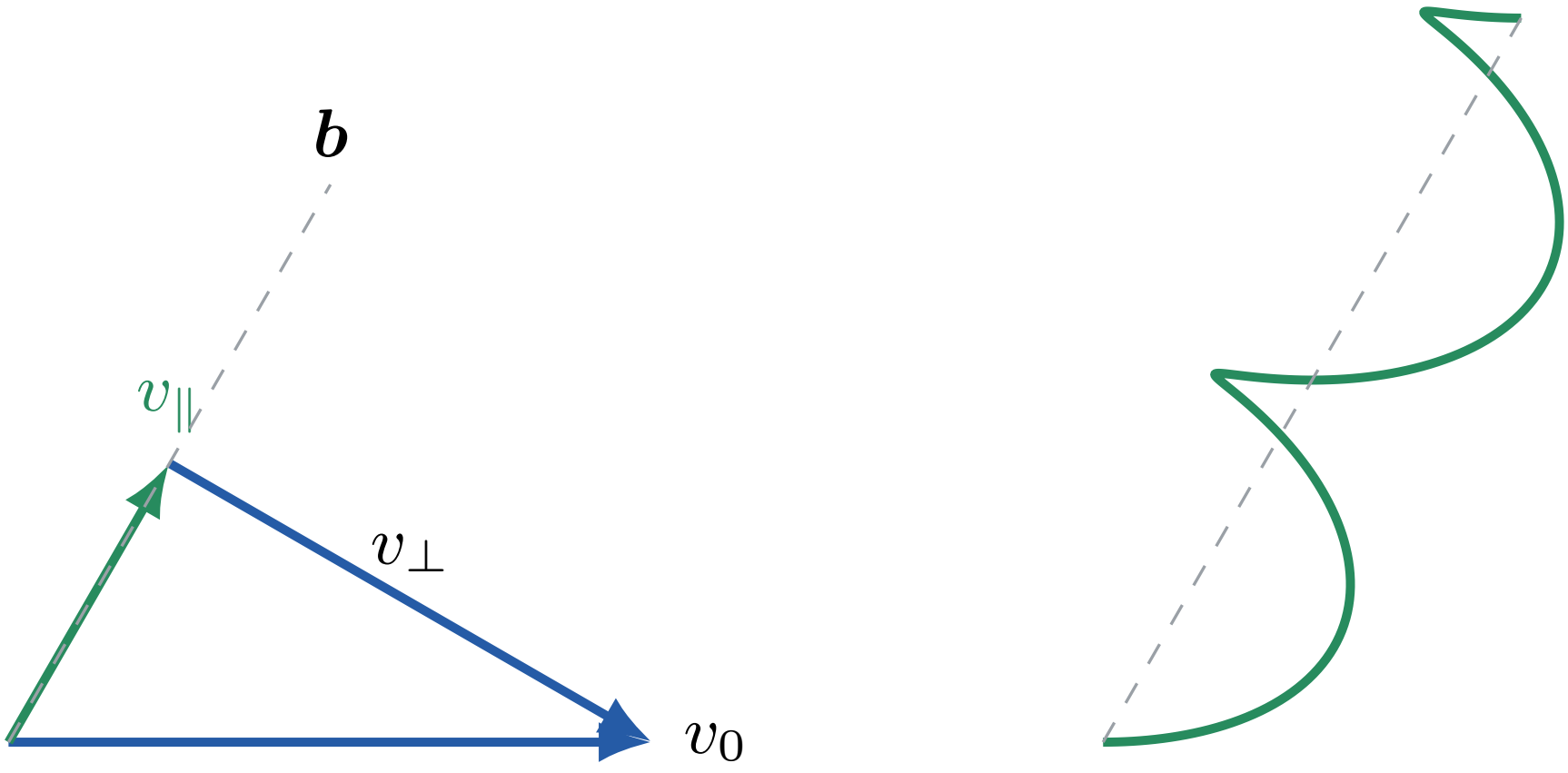

$$v_\parallel = v_0 \sin\theta, \quad v_\perp = v_0 \cos\theta$$

$$y = (v_0 \cos\theta/\Omega)(\cos\Omega t - 1)$$